

MUHAMMAD HAIKAL RAHMAN

Cybersecurity / SOC Analyst Portfolio | Master's Student in Informatics

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PROFESSIONAL SUMMARY

Master's student in Informatics at Institut Teknologi Bandung and Computer Science graduate with a five-month Telkomsel operations internship, ISC2 Certified in Cybersecurity (CC), practical blue-team training, and a completed Windows SOC home-lab core build. Hands-on with Windows Security telemetry, PowerShell logging, Sysmon, Splunk, detection engineering, cross-source incident investigation, false-positive analysis, MITRE ATT&CK; mapping, and technical documentation.

CYBERSECURITY PROJECTS

SOC Home Lab - Detection Engineering & Incident Investigation | Aug 2026

github.com/HaikalE/soc-home-lab

- Built an isolated Windows 11 Enterprise endpoint and Ubuntu Server SIEM environment in VMware; forwarded Windows Security, PowerShell Operational, and Sysmon telemetry to Splunk Enterprise.
- Engineered and validated DET-001 Multiple Failed Logons, DET-002 Encoded PowerShell Execution, and DET-003 PowerShell Spawned Command Shell.
- Investigated three controlled incidents by correlating Windows Security Event IDs 4625/4688, PowerShell 4104, and Sysmon Event ID 1; documented evidence, timelines, severity, disposition, and false-positive context.
- Reconstructed process lineage using ProcessGuid/ParentProcessGuid and analyzed authentication Status/Sub-Status to distinguish unknown-account events from valid-account wrong-password events.
- Mapped observed lab behavior to MITRE ATT&CK; and published detection notes, incident write-ups, architecture documentation, and final reports. All activity was performed in a controlled personal lab.

Secure Instant Messaging with AES-256 and RSA CRT | Jul 2024 - Oct 2024

- Built an Android messaging prototype using AES-256 for message encryption and RSA CRT for public-key operations with Kotlin, Node.js, and a NoSQL database; the security design supported undergraduate thesis and research work.

WORK EXPERIENCE

Teaching Assistant - Universitas Sumatera Utara, Fasilkom-TI | Aug 2024 - Present

- Support practical sessions and tutorials for Algorithm and Cryptography courses; prepare materials, review assignments, and mentor students.
- Contributed to faculty research activities that led to two registered software copyrights and a peer-reviewed IEEE conference paper.

Data Engineer Intern - PT Telekomunikasi Selular (Telkomsel) | Feb 2024 - Jun 2024

- Developed a Telegram ticketing bot to standardize network-quality support requests and support operational follow-up.
- Created Power BI dashboards for network-performance reporting and collaborated on recurring data-quality and reporting requirements.

CERTIFICATIONS & PRACTICAL TRAINING

- **ISC2 Certified in Cybersecurity (CC)** - ISC2
- **Blue Team Junior Analyst Training Pathway** - Centri | Completed Aug 10, 2026 | Certificate ID 248770024

Practical entry-level pathway covering Open-Source Intelligence, Digital Forensics, Vulnerability Management, DarkWeb Operations, Threat Hunting, and Network Analysis.

- Information Technology Specialist in Python - Certiport | 2025

TECHNICAL SKILLS

Security Operations: Alert triage, log analysis, incident documentation, false-positive analysis, Windows Security logs, PowerShell logging, Sysmon, threat hunting fundamentals, vulnerability management fundamentals

SIEM & Detection: Splunk Enterprise, Splunk Universal Forwarder, SPL, detection engineering, correlation searches, MITRE ATT&CK-aligned; investigation

Systems & Networking: Windows 11, Ubuntu Server, VMware Workstation, TCP/IP, DNS, HTTP/HTTPS fundamentals

Programming & Data: Python, SQL, JavaScript, Kotlin, Node.js, PostgreSQL, Power BI, Git, Docker, AWS S3

EDUCATION

Institut Teknologi Bandung (ITB) | Master of Informatics | Aug 2025 - Present

Relevant coursework: System Security and Privacy; Data and Software Security; Data Management; Programming for Data Analytics; Research Methodology.

Universitas Sumatera Utara | Bachelor of Computer Science | Aug 2021 - Feb 2025

GPA 3.88/4.00 | Thesis: Instant Messaging Security Using AES-256 and RSA CRT Algorithm.

Universiti Teknologi MARA | Study Abroad, Computer Science and Mathematics | Mar - Aug 2023

GPA 3.53/4.00 | Shah Alam, Malaysia

PUBLICATION, INTELLECTUAL PROPERTY & RECOGNITION

- IEEE ICERA 2025 paper: Instant Messaging Security Using Affine Cipher and RSA CRT Algorithm, pp. 30-35. DOI: 10.1109/ICERA66156.2025.11087351.
- Registered software copyrights: Somsuk-RSA application (EC00202510303) and Phony-RSA application (EC002025039293 / 000879554), Republic of Indonesia, 2025.
- Top 33 Programmer Talent ("Hacker" Track), IndonesiaNEXT 9 - Telkomsel | 2025.